

Cluster Sampling with Equal Probability Learning Activity

STA304 — Summer 2026

Learning Activity 7 Overview

In this activity you will apply statistical concepts from lecture 7 related to cluster estimation

Exercise

1. An inspector samples cans from a truckload of canned creamed corn to estimate the total number of worm fragments in the truckload. The truck has 580 cases; each case contains 24 cans. The inspector samples 12 cases at random, and subsamples 3 cans randomly from each selected case.

	Case											
	1	2	3	4	5	6	7	8	9	10	11	12
Can 1	1	4	0	3	4	0	5	3	7	3	4	0
Can 2	5	2	1	6	9	7	5	0	3	1	7	0
Can 3	7	4	2	6	8	3	1	2	5	4	9	0

- a) Is this a one-stage or two-stage cluster sampling design? Why?
- b) What are the PSUs in this problem? How many are there in the population? How many are there in your sample?

c) Estimate the total number of worm fragments, along with the standard error. Show your work.

d) Estimate the mean number of worm fragments, along with the standard error. Did you need to use the ratio estimator here? Show your work.

- e) (Optional) Estimate the mean and the total number of worm fragments in R using the survey package and ensure that it matches with c) and d). You can copy paste your **code and output** (like I did in the slides). You can alternatively upload a separate compiled Rmd.